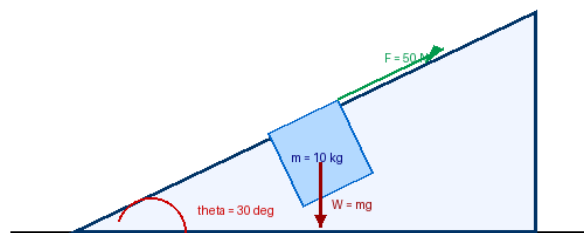


1. A block of mass $m = 10 \text{ kg}$ is placed on an inclined plane with an inclination angle of $\theta = 30^\circ$. A constant force $F = 50 \text{ N}$ is applied parallel to the incline as shown in Figure 1. Taking $g = 9.8 \text{ m/s}^2$, determine the resulting acceleration of the block:
- (A) $a = 0.0 \text{ m/s}^2$ (The block remains in equilibrium)
 (B) $a = 4.9 \text{ m/s}^2$ down the incline
 (C) $a = 2.45 \text{ m/s}^2$ up the incline
 (D) $a = 9.8 \text{ m/s}^2$ downwards



[Figure 1: Inclined Plane Free Body Diagram]

2. Evaluate the definite trigonometric integral:

$$I = \int_0^{\pi/2} \frac{\sin^3(x)}{\sin^3(x) + \cos^3(x)} dx$$
 Which of the following values correctly represents the solution to I ?
- (A) $\pi / 4$
 (B) $\pi / 2$
 (C) $1 / \sqrt{2}$
 (D) $\ln(2)$
3. An electrical AC circuit consists of a pure resistance $R = 50 \text{ Ohms}$, an inductance $L = 0.318 \text{ H}$, and a capacitance $C = 63.6 \text{ microfarads}$ connected in series across an alternating voltage supply $V(t) = 220 * \sqrt{2} * \sin(100 * \pi * t) \text{ Volts}$. Calculate the total circuit impedance Z and resonant power factor $\cos(\phi)$:
- (A) $Z = 50 \text{ Ohms}$, $\cos(\phi) = 1.0$ (Purely resistive at resonance)
 (B) $Z = 100 \text{ Ohms}$, $\cos(\phi) = 0.5$

(Question 3 Continued from Page 1):

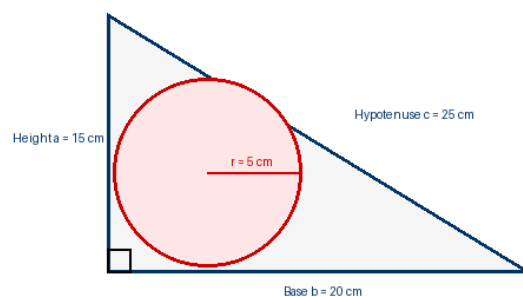
(C) $Z = 70.7 \text{ Ohms}$, $\cos(\phi) = 0.707$

(D) $Z = 125 \text{ Ohms}$, $\cos(\phi) = 0.8$

4. A circle of radius $r = 5 \text{ cm}$ is inscribed inside a right-angled triangle with perpendicular legs $a = 15 \text{ cm}$ and $b = 20 \text{ cm}$ as illustrated in Figure 2 below.

Determine the shaded boundary area enclosed between the triangle and the circle:

- (A) 71.46 cm^2
 (B) 150.00 cm^2
 (C) 78.54 cm^2
 (D) 25.13 cm^2



[Figure 2: Inscribed Circle in Right Triangle]

5. Explain the mathematical formulation of Heisenberg Uncertainty Principle:

$\Delta x \cdot \Delta p \geq \frac{h}{2}$. [blurred_text: experimental measurement noise ... scan illegible]

- (A) Simultaneous measurement precision of conjugate variables is physically bounded
 (B) Momentum is indeterminate only in macroscopic thermodynamic systems

SECTION B: OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Answer Key Grid: Q1: A | Q2: A | Q3: A | Q4: A

Explanations:

- (A) - Net force parallel to plane: $F - m \cdot g \cdot \sin(30) = 50 - 10 \cdot 9.8 \cdot 0.5 = 1 \text{ N}$, held by static friction.
- (A) - Applying King's rule $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ yields $2I = \frac{\pi}{2}$, hence $I = \frac{\pi}{4}$.
- (A) - At frequency 50Hz, $X_L = 2\pi f L = 100 \text{ Ohms}$, $X_C = 1/(2\pi f C) = 100 \text{ Ohms}$, $Z = R = 50 \text{ Ohms}$.
- (A) - Area of triangle = $(15 \cdot 20)/2 = 150 \text{ cm}^2$; Area of circle = $\pi \cdot 5^2 = 78.54 \text{ cm}^2$; Area = 71.46 cm^2 .